

Renaldo Miya

Mechanical Engineering at the University of Waterloo

Portfolio: renaldomiya.github.io

Email: renaldo.miya@uwaterloo.ca

Linkedin: linkedin.com/in/renaldo-miya/

EXPERIENCE

TRIUMF – ARIEL Engineering Assistant (January 2026 – April 2026)

- Designed a rectangular stainless steel support frame in SolidWorks for a radioactive gas pumping manifold, producing a rigid, minimal-footprint enclosure mounted directly to vacuum package room walls.
- Optimized member sizing and clamp layout through iterative ANSYS FEA, satisfying 2 mm deflection and 105 MPa stress limits while minimizing total parts in a radiation-exposed assembly.
- Applied DFM/DFA principles to position every support and clamp for full technician access to gauges, sensors, and KF connections, enabling independent component removal during maintenance.
- Produced detailed manufacturing drawings with GD&T callouts for fabrication and inspection, reducing fastener count and procurement complexity to streamline assembly in the ARIEL Target Hall.
- Generated and managed BOMs within a PDM system, coordinating with engineers and technicians to maintain configuration control and design traceability across iterative frame revisions.
- Supported fabrication using manual machining tools, validating tolerances and resolving fit-up issues during assembly to ensure CAD design intent translated cleanly into the installed structure.

BC Hydro – Mechanical Engineering Co-op (May 2024 – August 2024)

- Reviewed and redlined P&ID drawings and mechanical layouts, performing design evaluations and recommending modifications to improve constructability and compliance with standards.
- Performed on-site field inspections to confirm installation accuracy, compliance, and overall quality; recorded detailed findings, and supported the implementation of design modifications and improvements.
- Translated RFIs and Field Change Forms into engineered solutions by coordinating with multidisciplinary teams, assessing technical constraints, and issuing practical design revisions to support construction progress.

UW Formula Electric – Mechanical Design & Composites Member (September 2023 – December 2023)

- Designed and modeled a lightweight relay enclosure in SolidWorks by incorporating DFM/DFA principles, which reduced overall weight while simplifying assembly for faster integration into the vehicle.
 - Assisted with composite manufacturing of the undertray by performing mold preparation, body filling, primer application, and sanding, which ensured high-quality surface finish and structural integrity.
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PROJECTS

UW Robotics – Robotic Claw for University Rover Challenge

- Designed and engineered a motor-driven rack-and-pinion robotic claw capable of grasping and lifting objects up to 5 kg, owning the project from concept through functional prototype.
 - Modeled complete assemblies and detailed components in SolidWorks, translating rotational motor output into precise linear jaw actuation through a custom gear-driven mechanism.
 - Integrated 3D-printed structural components with machined aluminum parts, balancing weight, manufacturability, and structural stiffness to prevent deflection under peak load.
 - Applied design-for-manufacturability and assembly principles to simplify fabrication and improve serviceability of motors, fasteners, and power transmission components.
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EDUCATION

University of Waterloo – Bachelor of Mechanical Engineering (Honors), Class of 2029

- Recipient of the President's Entrance Scholarship